

SubMagic Avatar POC

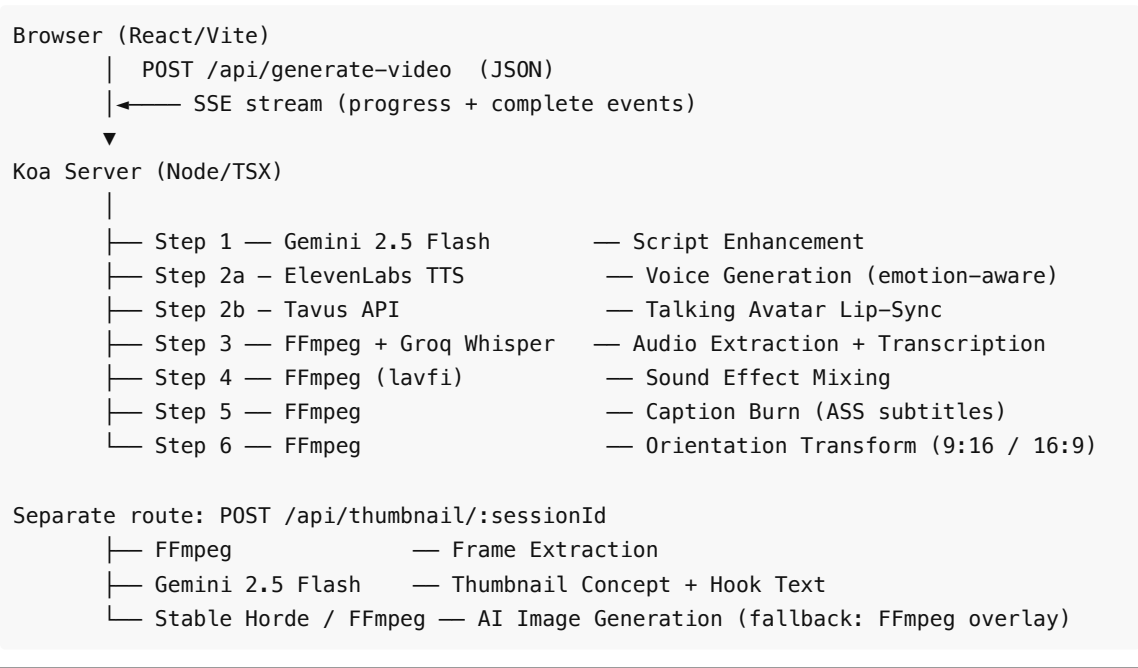
Overview

A proof-of-concept pipeline that turns a text script into a fully produced short-form video featuring a talking avatar — with AI-enhanced narration, lip-synced video, word-level captions, optional sound effects, and an AI-generated thumbnail.

Stack: Vite + React 19 (frontend) · Koa (backend) · TypeScript end-to-end

Delivery: Server-Sent Events (SSE) for real-time progress streaming

Architecture



ElevenLabs — Complete Configuration Reference

ElevenLabs is the **voice layer** of the image-to-video pipeline. It converts the AI-enhanced script into a natural-sounding MP3 that Tavus uses to drive the avatar's lip movement.

Plans & Limits

Feature	Free	Starter (\$5/mo)	Creator (\$22/mo)	Pro (\$99/mo)	Scale (\$330/mo)
Characters / month	10,000	30,000	100,000	500,000	2,000,000
Premade voices via API	Yes	Yes	Yes	Yes	Yes
Voice Library (community) via API	No	Yes	Yes	Yes	Yes

Instant Voice Cloning	No	Yes (3 voices)	Yes (30 voices)	Yes (160 voices)	Yes (660 voices)
Professional Voice Cloning	No	No	No	Yes	Yes
Voice Design (AI-generated voice)	No	No	Yes	Yes	Yes
Projects (long-form narration)	No	No	Yes	Yes	Yes
Dubbing Studio	No	No	Yes	Yes	Yes
Commercial license	No	Yes	Yes	Yes	Yes
Priority queue	No	No	No	Yes	Yes

Free plan API restriction: Voices added from the Voice Library (Adam, Rachel, Bella, Josh, etc.) cannot be used via API. Only ElevenLabs' own built-in "premade" voices work.

Configurable Parameters (Text-to-Speech API)

Voice Selection

Parameter	Description	Free	Paid
voice_id	Which voice to use	Premade only	Any voice
Premade voices	Built-in ElevenLabs voices (Brian, Alice, Eric, Sarah, George, Charlotte, Laura, Matilda, etc.)	Yes	Yes
Voice Library voices	Community-contributed voices	No	Yes
Cloned voices	Your own uploaded voice	No	Yes (Starter+)
Professional clone	High-fidelity custom voice	No	Yes (Pro+)

Voice Settings (per request)

Parameter	Range	Description	Effect on delivery
stability	0.0 – 1.0	Consistency of tone	0 = very expressive and variable · 1 = flat and robotic
similarity_boost	0.0 – 1.0	Adherence to original voice character	Higher = closer to the base voice but less flexible
style	0.0 – 1.0	Style exaggeration multiplier	Amplifies the voice's inherent emotional character

use_speaker_boost	true / false	Enhances clarity and presence	Recommended for most voices; disable for softer emotions (sad)
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How this POC maps emotion to voice settings

Emotion	stability	style	similarity_boost	use_speaker_boost
Neutral	0.71	0.00	0.50	Yes
Happy	0.50	0.30	0.75	Yes
Excited	0.30	0.55	0.75	Yes
Motivational	0.38	0.48	0.75	Yes
Professional	0.82	0.00	0.80	Yes
Sad	0.65	0.20	0.65	No

Model Selection

Model	Quality	Latency	Languages	Free
eleven_multilingual_v2	Highest	Medium	29	Yes
eleven_turbo_v2_5	High	Fast	32	Yes
eleven_turbo_v2	High	Fast	13	Yes
eleven_flash_v2_5	Good	Very fast	32	Yes
eleven_monolingual_v1	Good	Medium	English only	Yes

This POC uses `eleven_multilingual_v2` for best quality. Switching to `eleven_turbo_v2_5` reduces latency with minimal quality loss.

Output Format

Parameter	Options	Default	Notes
output_format	mp3_44100_128, mp3_44100_192, pcm_16000, pcm_22050, pcm_24000, ulaw_8000	mp3_44100_128	Higher bitrate = better audio quality for lip-sync
Response	Binary audio stream	–	Returned directly as audio/mpeg bytes

Voice Mapping in This POC

The POC resolves voices dynamically at runtime via `GET /v1/voices`, then matches by name preference — so it works on free and paid plans without hardcoded IDs.

UI Option	Preferred Voices (tried in order)	Character
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Professional Male	Brian → George → Daniel → Eric → Liam	Deep, authoritative American/British male
Professional Female	Alice → Charlotte → Laura → Matilda	Clear, polished British/American female
Friendly Female	Sarah → Jessica → Lily → River	Warm, soft American female
Motivational Male	Eric → Brian → Chris → Will	Confident, energetic American male

If a preferred name is not found in the account, the next candidate is tried. The first available voice is used as a hard fallback so the pipeline never fails due to missing voice.

How ElevenLabs Audio Reaches Tavus

1. ElevenLabs MP3 is saved to `tmp/{sessionId}/tts.mp3`
2. Koa server exposes it at `GET /api/audio/{sessionId}`
3. If `SERVER_BASE_URL` is configured → `audio_url = SERVER_BASE_URL/api/audio/{sessionId}` is passed to Tavus
4. Tavus lip-syncs the avatar to the ElevenLabs recording instead of generating its own voice
5. Without `SERVER_BASE_URL` (local dev without tunnel) → Tavus falls back to its own TTS

ElevenLabs — Image to Video

Concept

"Image to Video" is the workflow of combining a **static avatar image** with an **ElevenLabs audio track** to produce a **talking head video**. ElevenLabs provides the voice; a lip-sync engine drives the avatar's mouth movements from the audio waveform.



Why ElevenLabs produces better image-to-video results

Dimension	macOS say / generic TTS	ElevenLabs
Voice quality	Robotic, synthetic	Human-quality, natural prosody
Emotion control	None	Full parameter control per request
Phoneme clarity	Poor (blurry mouth sync)	Sharp phoneme boundaries → precise lip movement
Language support	English only (macOS)	29–32 languages
Voice variety	4–6 system voices	100s of voices, custom cloning

Commercial use	Mac-only, non-commercial	Licensed for commercial use (Starter+)
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Full Pipeline: Static Image → Talking Video

Step	Service	Input	Output
1	Gemini 2.5 Flash	Raw script + emotion + language	Polished, length-targeted script
2a	ElevenLabs	Script + voice option + emotion	MP3 audio with expressive delivery
2b	Tavus	Avatar replica ID + ElevenLabs MP3 URL	Talking avatar MP4
3	Groq Whisper	Avatar MP4 audio	Word-level timestamps
4	FFmpeg	Avatar MP4 + SFX at timestamps	SFX-mixed MP4
5	FFmpeg	SFX MP4 + ASS subtitle file	Captioned MP4
6	FFmpeg	Captioned MP4 + orientation setting	Final portrait / landscape MP4

Alternative Lip-Sync Backends

ElevenLabs audio output is backend-agnostic. The same MP3 feeds into any lip-sync engine:

Service	Type	ElevenLabs integration	Notes
Tavus (current)	Cloud API	audio_url parameter	High realism, managed cloud
fal.ai OmniHuman	Cloud API	audio_url parameter	Very high realism, accepts audio URL
SadTalker	Local model	Local file path	Python + GPU required
LivePortrait	Local model	Local file path	Faster, portrait-optimised
Wav2Lip	Local model	Local file path	Widely supported, older
HeyGen	Cloud API	Upload audio	Commercial product

Tavus — Complete Configuration Reference

Tavus is the **lip-sync and video rendering layer**. It takes a static avatar replica and either a text script or a pre-generated audio file, renders the avatar speaking, and returns a downloadable MP4.

Plans & Limits

Feature	Free / Sandbox	Growth (\$47/mo)	Scale (custom)	Enterprise (custom)
Video minutes / month	~5 min (limited)	25 min	Custom	Custom

Watermark	Always on	No	No	No
Custom replicas	No (use stock)	Yes	Yes	Yes
Stock replicas	Yes	Yes	Yes	Yes
Video resolution	720p	1080p	1080p+	4K
Background customisation	No	Yes	Yes	Yes
audio_url (BYOA)	Limited	Yes	Yes	Yes
Greenscreen removal	No	Yes	Yes	Yes
Async webhook	No	Yes	Yes	Yes
API rate limit	Low	Medium	High	Custom SLA
Commercial license	No	Yes	Yes	Yes

Sandbox restriction: Free tier videos always include a Tavus watermark and output at 720p. `disable_watermark` in `properties` is a paid-plan-only feature.

Configurable Parameters (POST /v2/videos)

Core Fields

Parameter	Type	Required	Free	Paid	Description
replica_id	string	Yes	Stock only	Any replica	ID of the avatar replica to animate
script	string	Yes (or audio_url)	Yes	Yes	Text the avatar will speak (Tavus runs its own TTS)
audio_url	string	Yes (or script)	Limited	Yes	Public URL to a pre-generated audio file (e.g. ElevenLabs MP3)
video_name	string	No	Yes	Yes	Label for the generated video

Visual Customisation

Parameter	Type	Free	Paid	Description
background_url	string	No	Yes	URL to a background image or video rendered behind the avatar
background_color	string	No	Yes	Hex or RGBA fallback background color (e.g. #1a1a2e)

Properties Object

Property	Type	Free	Paid	Description
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apply_greenscreen	boolean	No	Yes	Remove the green screen from the replica video
disable_watermark	boolean	No	Yes	Remove the Tavus watermark from the output
max_duration	number (seconds)	No	Yes	Cap the output video duration
crop_to_aspect_ratio	boolean	No	Yes	Crop the output to a specific aspect ratio

Language & Voice

Parameter	Type	Free	Paid	Description
language	string (ISO 639-1)	Limited	Yes	Language code for Tavus's internal TTS (e.g. en, es, fr). Ignored when audio_url is provided.

Replica Configuration (separate API, before video generation)

Feature	Free	Paid	Notes
Use stock replicas	Yes	Yes	Pre-built avatars provided by Tavus
Create custom replica	No	Yes	Upload video of yourself to train a replica
Replica voice	Fixed (tied to replica)	Fixed	Voice is baked into the replica; use audio_url to override
Replica appearance	Fixed	Fixed	Visual is baked into the replica
Green screen replica	No	Yes	Replica trained against green screen for background replacement

What This POC Exposes as UI Options

UI Field	Maps to Tavus param	Free	Paid
Custom Replica ID	replica_id override	Stock IDs only	Any replica
Background URL	background_url	—	Yes
Background Color	background_color	—	Yes
Language	language	Limited	Yes
Apply Greenscreen	properties.apply_greenscreen	—	Yes
Disable Watermark	properties.disable_watermark	—	Yes

Video Output

Property	Free / Sandbox	Paid
Resolution	720p	1080p (Growth), 4K (Enterprise)
Format	MP4 (H.264)	MP4 (H.264 / H.265)
Watermark	Yes, always	No (with <code>disable_watermark: true</code>)
Delivery	Polling (GET <code>/v2/videos/{id}</code>)	Polling or webhook
Download URL	<code>download_url</code> in status response	Same
Max video duration	~60s (sandbox)	5 min (Growth), unlimited (Scale+)

Polling Flow (Current Implementation)

```
POST /v2/videos → { video_id }
├─ poll GET /v2/videos/{video_id} every 6s
│   ├── status: "queued" → keep polling
│   ├── status: "processing" → keep polling (log elapsed time)
│   ├── status: "ready" → download_url available → download MP4
│   └─ status: "failed" → throw error with message
```

Timeout: 10 minutes. Real-world average render time: 60–180 seconds depending on script length and plan tier.

Combined Free-Tier Capability Summary

Feature	ElevenLabs Free	Tavus Free/Sandbox
Voice quality	Human-quality (premade voices)	Replica's own voice (stock replicas)
Emotion control	Full (stability/style params)	None (Tavus TTS ignores emotion)
Custom voice	No	No
Background	N/A (audio only)	No
Watermark	N/A	Yes, always
Output resolution	N/A	720p
Commercial use	No	No
Monthly limit	10,000 chars (~25 videos of 30s)	~5 min of video
Languages	29	Limited

Recommended free-tier workflow: Use ElevenLabs for voice (emotion-tuned audio) + Tavus stock replica for lip-sync. Accept the watermark and 720p output. Upgrade Tavus Growth (\$47) to remove watermark and get 1080p for production use.

Pipeline Steps (Full)

Step 1 — Script Enhancement (Gemini 2.5 Flash)

Raw script → Gemini with emotion + duration target (15s/30s/60s/90s = 37/75/150/225 words). Gemini rewrites for correct length and tone. Falls back to original script on 503.

Step 2a — Voice Generation (ElevenLabs)

Script → ElevenLabs TTS → MP3 with emotion-tuned voice settings. Saved to session dir. If `SERVER_BASE_URL` is set, URL passed to Tavus. Falls back to Tavus TTS if ElevenLabs fails.

Step 2b — Talking Avatar (Tavus)

Creates video via `POST /v2/videos` with all configured Tavus options. Polls every 6s until ready, then downloads MP4.

Step 3 — Transcription (Groq Whisper)

Audio extracted from avatar MP4 (WAV). `whisper-large-v3` transcribes with word-level timestamps for captions + SFX alignment.

Step 4 — Sound Effects (FFmpeg lavfi)

Three synthesised effects (pop/ding/whoosh) mixed at user-selected word timestamps at 40% volume via `amix`.

Step 5 — Caption Burn (FFmpeg + ASS)

Word timestamps → `.ass` subtitle file → burned into video. Three style presets (Viral, Professional, Creator). Fully customisable per session.

Step 6 — Orientation Transform (FFmpeg)

Landscape: stream copy. Portrait (9:16): scale to 1080px wide, pad to 1080×1920 with black bars.

Thumbnail Generation

1. Frame extracted from `avatar.mp4` at style-specific position
2. Gemini Vision: frame + script → `hookLine`, `subLine`, Stable Diffusion prompt
3. Stable Horde AI image (free, anonymous, 30s timeout) → fallback FFmpeg overlay

Five styles: Engaging, Normal, Cinematic, Professional, Dynamic.

AI Token Usage

Tracked per Gemini call, displayed in UI after generation.

Call	Returns	Typical input	Typical output
<code>enhanceScript</code>	<code>{ text, tokens }</code>	300–600	75–250
<code>generateConcept (thumbnail)</code>	logged only	800–1,500	200–400

ElevenLabs bills by characters (not tokens). Groq Whisper and Tavus bill by audio/video duration.

Services & APIs

Service	Role	Free Tier	Auth
ElevenLabs	TTS with emotion	10k chars/mo, premade voices	ELEVENLABS_API_KEY
Tavus	Lip-sync + video	~5 min, watermark, 720p	TAVUS_API_KEY
Gemini 2.5 Flash	Script + thumbnail concept	Free tier available	GEMINI_API_KEY
Groq Whisper	Transcription + timestamps	Free tier	GROQ_API_KEY
Stable Horde	Thumbnail AI image	Free, anonymous	None
FFmpeg	Audio/video processing	Free, local	None

Environment Variables

```
ELEVENLABS_API_KEY=      # ElevenLabs – free plan works for premade voices
GEMINI_API_KEY=          # Google AI Studio
GROQ_API_KEY=            # Groq
TAVUS_API_KEY=           # Tavus
TAVUS_REPLICA_ID=        # fallback replica ID for all avatars
TAVUS_REPLICA_ID_AVATAR1= # per-avatar overrides (optional)
TAVUS_REPLICA_ID_AVATAR2=
TAVUS_REPLICA_ID_AVATAR3=
TAVUS_REPLICA_ID_AVATAR4=

# Required for ElevenLabs audio → Tavus lip-sync
# Tavus must fetch the MP3 from a public URL.
# Local dev: npx localtunnel --port 3001 → SERVER_BASE_URL=https://xxx.loca.lt
SERVER_BASE_URL=
```

Running Locally

```
npm install
npm run dev      # Vite on :5173 + Koa on :3001

# Enable ElevenLabs lip-sync locally:
npx localtunnel --port 3001
# → add SERVER_BASE_URL=https://xxx.loca.lt to .env
```

Known Limitations

- **SERVER_BASE_URL** needed for ElevenLabs → Tavus lip-sync. Without it Tavus uses its own TTS.
- **ElevenLabs free = 10k chars/month** — ~25 videos of 30s each.
- **Tavus free always watermarks** — upgrade to Growth (\$47/mo) for clean output.
- **Tavus latency** — 60–180s render time, pure polling (no webhooks yet).
- **Single-user sessions** — no auth, files deleted on "Generate Another".
- **Stable Horde** — variable queue times; FFmpeg fallback always available.
- **analyzeSoundEffects** — Gemini auto-SFX is implemented but not wired; sound effects are user-selected only.